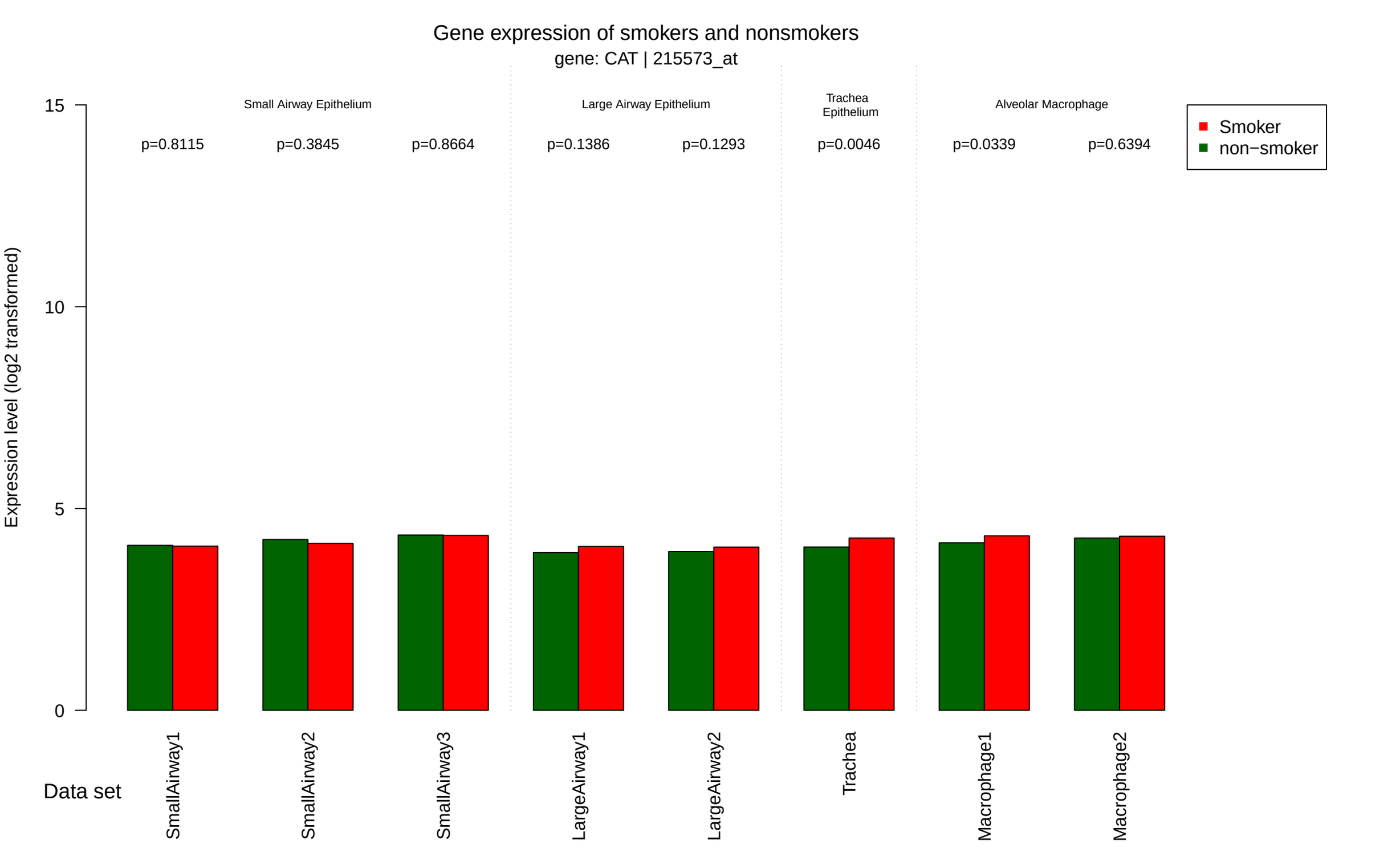
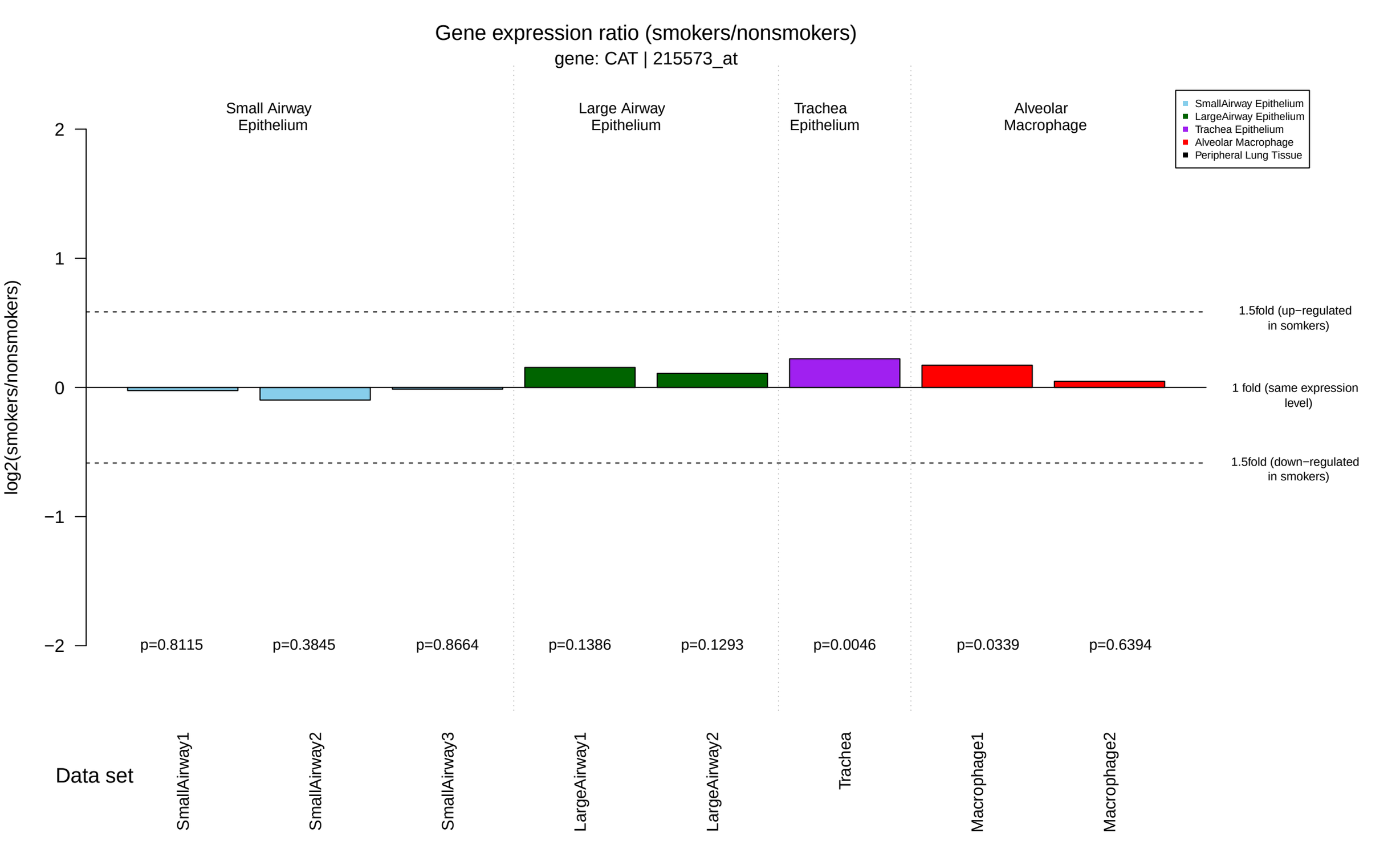
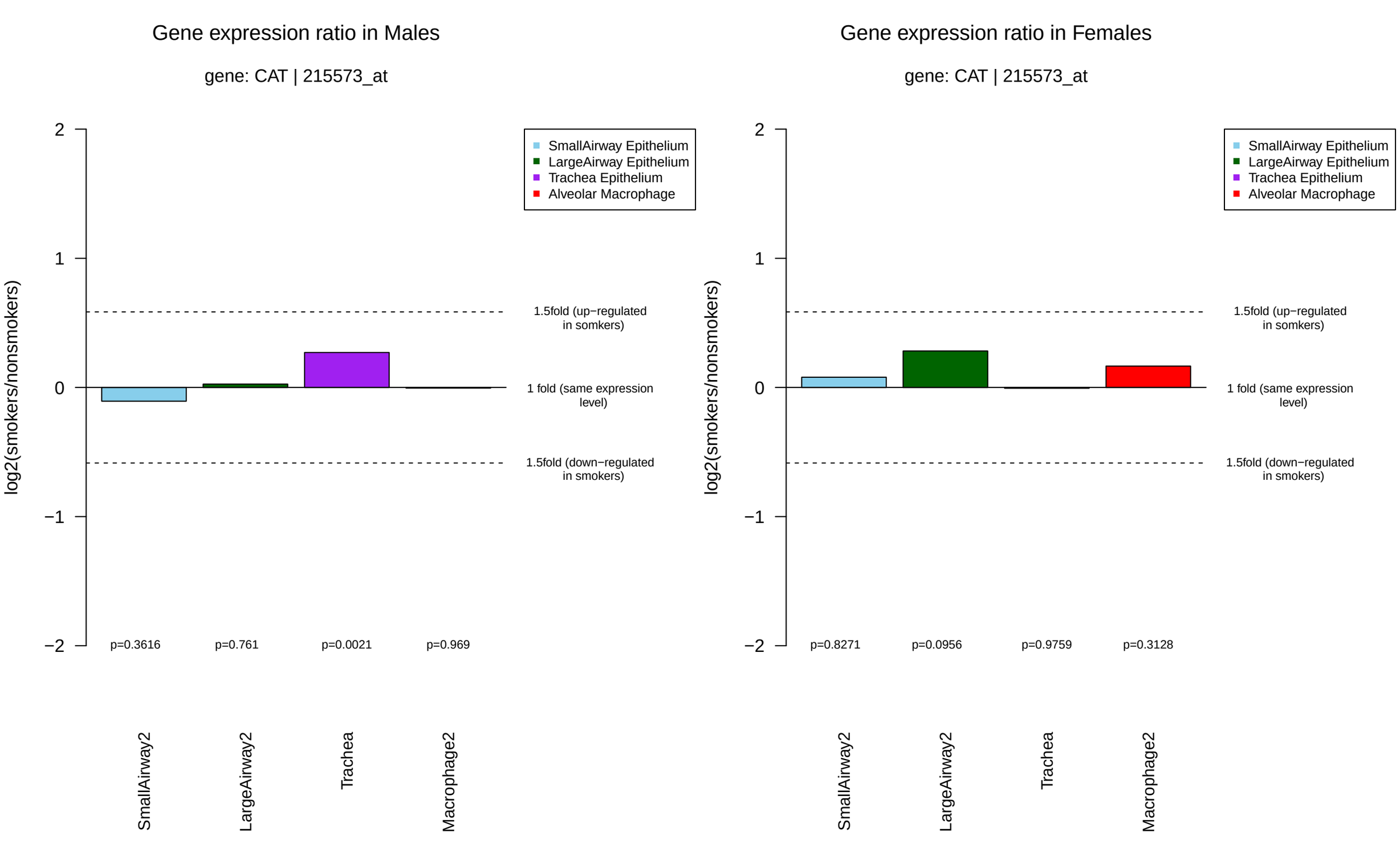
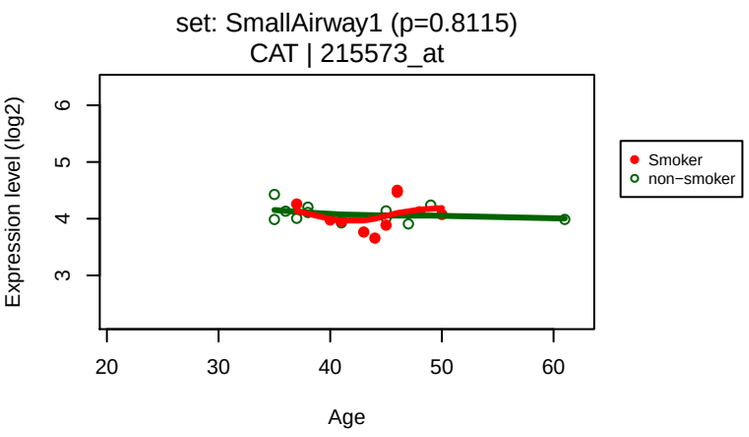
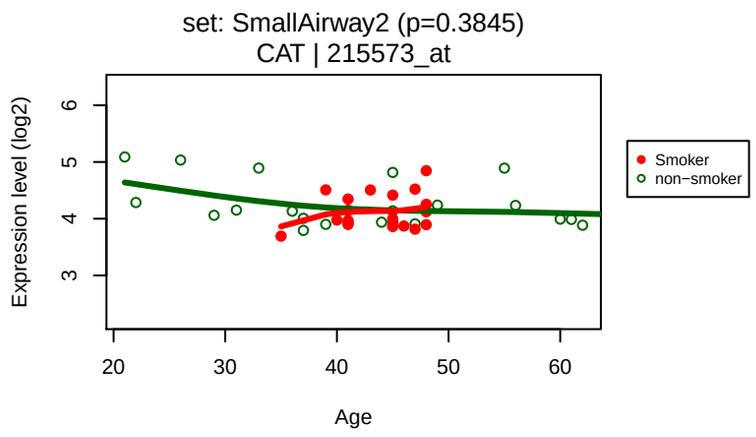
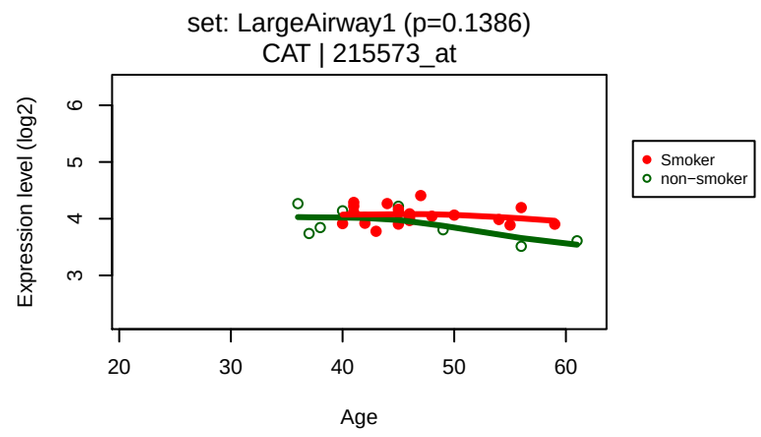

Gene expression of smokers and nonsmokers
gene: CAT | 215573_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8115
p=0.3845
p=0.8664
p=0.1386
p=0.1293
p=0.0046
p=0.0339
p=0.6394
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

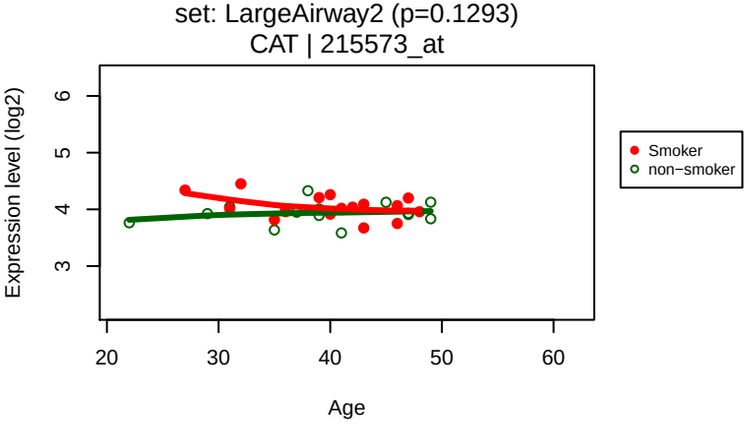

Gene expression ratio (smokers/nonsmokers)
gene: CAT | 215573_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8115
p=0.3845
p=0.8664
p=0.1386
p=0.1293
p=0.0046
p=0.0339
p=0.6394
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

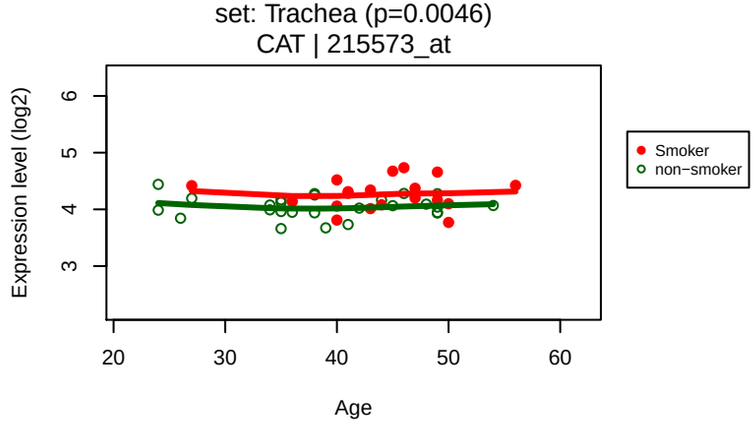

Gene expression ratio in Males
gene: CAT | 215573_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3616
p=0.761
p=0.0021
p=0.969
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CAT | 215573_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8271
p=0.0956
p=0.9759
p=0.3128
SmallAirway2
LargeAirway2
Trachea
Macrophage2

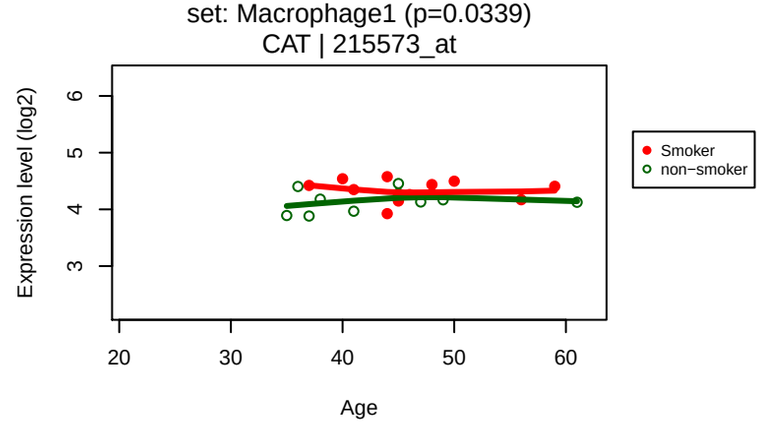

set: SmallAirway1 (p=0.8115)
CAT | 215573_at
Expression level (log2)
Age
Smoker
non−smoker

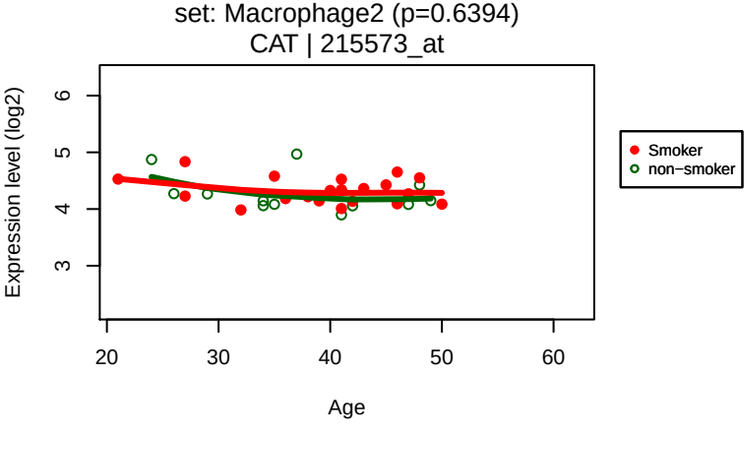

set: SmallAirway2 (p=0.3845)
CAT | 215573_at
Expression level (log2)
Age
Smoker
non−smoker

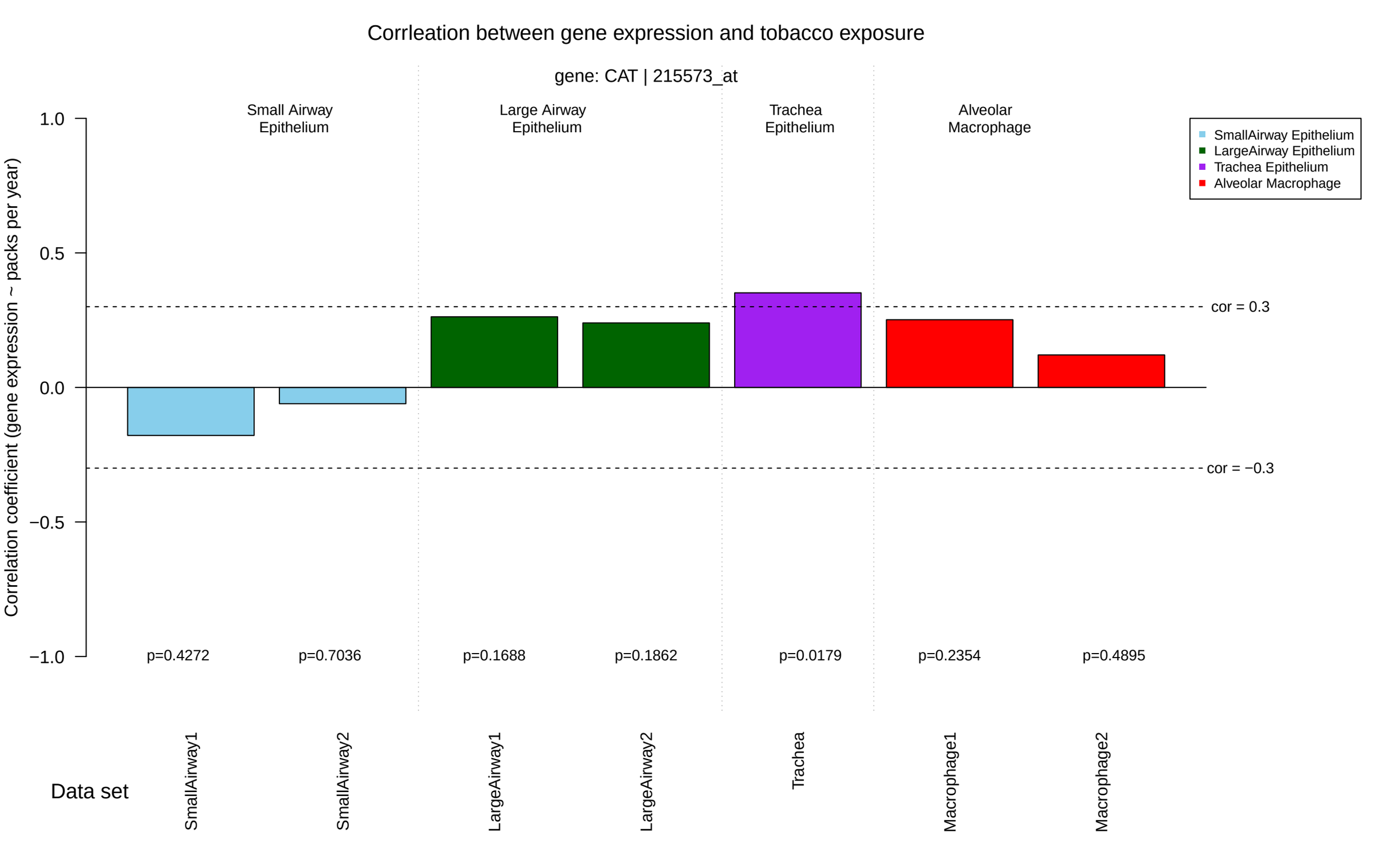

set: LargeAirway1 (p=0.1386)
CAT | 215573_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1293)
CAT | 215573_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0046)
CAT | 215573_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0339)
CAT | 215573_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6394)
CAT | 215573_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CAT | 215573_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4272
p=0.7036
p=0.1688
p=0.1862
p=0.0179
p=0.2354
p=0.4895
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2